Assessment of Cumulative Lifetime Seaway Loads for Ships

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ABSTRACT

This paper presents a reliability-based design methodology for primary seaway loadings on surface ships. The goal for such a design methodology is to have one consistent method for developing extreme primary hull girder loads for reliability-based strength design criteria as well as lifetime exceedance loads for reliability-based fatigue strength design criteria. The primary hull girder loads, i.e., vertical bending, lateral bending, and torsion, include contributions from both the low frequency (wave-induced) and the high frequency (slam-induced) domains for vertical and lateral bending. The contributions of secondary loads, such as hydrostatic and dynamic pressure loads, are important but are beyond the scope of this paper. A method is presented for calculating lifetime seaway loads as functions of ship and seaway parameters. The method is embodied in a PC based computer program called SPECTRA.

Primary Loads

Ships respond to the sea as if they were simple beams that are bent in the vertical plane (hogging and sagging), the lateral plane (athwartship bending), and twisted about their longitudinal axes (torsion). **Figure 1** shows these primary load cases. Two components to hull girder bending can be identified: a low frequency, wave-induced (wave buoyancy) loading and a high frequency, slam-induced whipping. This paper addresses both low and high frequency loadings individually as well as their combinations.

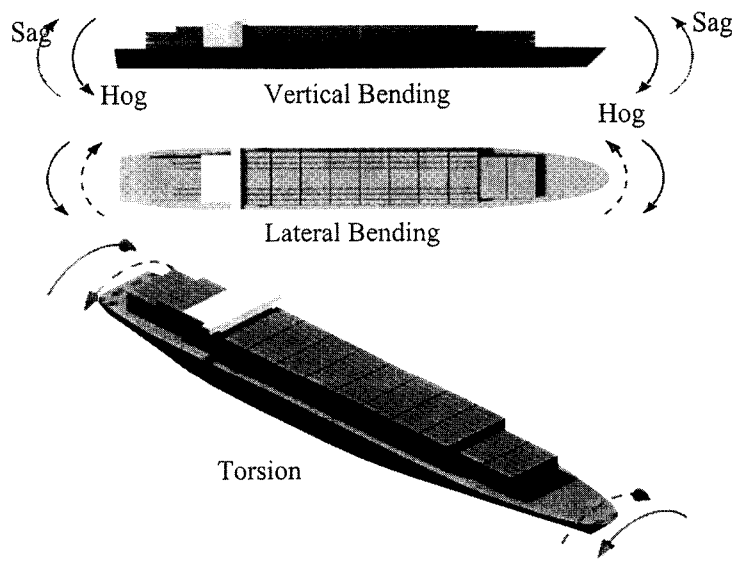


FIGURE 1: *Types of Hull Girder Loading*

Estimation of Extreme Loads

Modeling Lifetime Load Exposure

A method for estimating maximum, or extreme, seaway-induced moments acting on ships, as well as lifetime exceedance load spectra, has been developed at the Naval Surface Warfare Center, Carderock Division (NSWCCD) (Sikora 1983, 1986, and 1998). The ship's lifetime exposure at sea is divided into blocks or cells corresponding to its operational and environmental parameters. The size of the cells must be small enough so that the conditions within them can be considered statistically stationary, that is, the values of the statistical parameters describing the distribution do not change with time. A few thousand cells are typically needed to define lifetime conditions at sea.

The time in each of these statistically stationary cells is a function of ship heading and speed in addition to the wave height and average wave period in the ocean. The time spent in each cell is the product of the total

time at sea times the probability of being in that operating condition times the probability that the sea condition exists in the desired ocean area. The statistics of the ship response in each of these stochastic cells can be used to determine probability density functions. The combination of all of the probability density functions and their associated number of events is used to determine an extreme seaway loading with an associated reliability. The exceedance loads for fatigue are determined by selecting a set of loads and calculating the number of exceedance cycles from each of the probability distributions.

A naval architect selects several parameters to initiate the design or analysis: the lifetime the ship is to operate (T), the geographic area in the form of ocean statistics for the probability of occurrences of wave heights (Pwh_i) and modal periods (Pmp_i), and the probabilities of operating at particular speeds (Ps_i) and headings (Ph_i). These input parameters completely define the time (T_i) the ship will spend in each of the cells as follows:

$$T_i = T\, Pwh_i\, Pmp_i\, Ps_i\, Ph_i \qquad (1)$$

The number of loading events or cycles (N_i) in each cell corresponds to the number of waves encountered by the ship in that cell, and can be computed as

$$N_i = T_i\, \omega_{ei} / 2\pi \qquad (2)$$

where ω_{ei} is the average wave encounter frequency for that cell as provided by

$$\omega_{ei} = \left| \omega + \omega^2 (v/g) \cos(\theta) \right| \qquad (3)$$

where v is the ship speed, g is the gravitational acceleration (v/g is in units of 1/sec), and θ is the heading. The wave frequency (() is the modal frequency and can be approximated by:

$$\omega = 0.4 \sqrt{g/H} \qquad (4)$$

where H is the significant wave height for that cell.

WAVE-INDUCED LOADS

Time History of Loads

The calculation of the probability density functions for each cell is not so straightforward. One approach is to generate time histories of the load for each cell. This can be done by conducting model tests or through three dimensional, analytical computer codes. The time histories must be analyzed to determine the peak load at each wave event and the loads compiled into statistics for each cell. The cell statistics can then be curve fit into probability density functions through conventional distributions (Weibull distributions for example) or used directly as numerical distributions. However, this approach is not feasible for analyses typically involving several thousand cells. An extensive model test program would take on the order of one year to complete. Running analytical codes long enough to generate statistically significant time histories, analyzing the data, and then developing probability density functions could also take an inordinately long time.

Frequency Representation of Loads

Another approach to represent the probability density functions for the wave-induced (M) portion of the loads is the frequency domain instead of the time domain. The response of the ship in the frequency domain (commonly known as the response amplitude operators or RAOs) can be multiplied by a sea spectra of the wave energy to generate a load response spectrum in the frequency domain for each cell. Each response spectrum defines an amplitude probability distribution that can be described by a Rayleigh distribution. The RAOs for a given ship are ordinarily only functions of ship speed and heading and can be obtained from model tests, sea trials, Froude scaling from similar vessels, algorithms, or analytical computer programs. There are several, well known sea spectra that exist as closed-form equations and are only functions of wave height and modal period. This approach works for the large majority of cells in which there is no slam-induced whipping as well as for the high cycle-low load portion of the cells with whipping. This approach is also adequate

for torsion because there are no significant slam responses in torsion.

The quasi-static, wave-induced hull girder bending and torsion loads can be completely defined in the frequency domain. The lifetime operational domain of the ship can be divided into a number of stochastic cells (functions of sea state, heading, and speed). The response function for a cell is the product of the appropriate RAO and sea spectral description. The family of responses in each cell is a function of the response function and the number of lifetime events in that cell (N_i). Twice the area under each response function (i^{TH} cell) is the mean squared amplitude (E_i) of the responses within that cell. Since the peaks of the response function follow a Rayleigh distribution, the most probable maximum value ($Mmax_i$) for a cell is:

$$Mmax_i = [E_i \ln(N_i)]^{0.5} \quad (5)$$

Therefore, $Mmax_i$ would be exceeded once in N_i cycles.

The cumulative probability (CP_I, also referred to as the reliability) of load M for each cell is defined by:

$$CP_i = [1 - \exp(-Y)]^{Ni} \quad (6)$$

where $Y = M^2/E_i$ for a Rayleigh distribution. For the most probable maximum value of a cell, $Y = (Mmax_i)^2/E_i$, and the cumulative probability is 0.368 for all N. Since failure probability plus reliability equals unity, the occurrence probability for M_{max_i} is 0.632.

The net cumulative probability for load M for all of the cells is the product of the individual cumulative probabilities for each cell.

$$CP_{net} = \prod_{i=1,n} CP_i \quad (7)$$

Hence, one can iterate among values of M to arrive at any desired risk or reliability.

Since the Navy was not interested in quantitative reliability levels twenty years ago, an alternative method for predicting an extreme value for the wave-induced loads was developed. In Sikora, Dinsenbacher, and Beach (1983) and Sikora (1998), a maximum lifetime load ($Mmax_L$) corresponding to that load which is exceeded once by all of the wave-induced cells cumulatively was proposed and used.

$$\sum_i \exp[\,-(Mmax_L)^2 / E_i\,]\, N_i = 1.0$$

In practice, $Mmax_L$ is 10 to 15% greater than the $Mmax$ of the worst cell.

This maximum wave-induced moment was not meant to be used alone in these methodologies. A slam-induced whipping component was explicitly added to M_{max_L} to yield a combined lifetime maximum load (CM). For naval combatants, CM could be 30 to 50% larger than Mmax. In this method, $Mmax_L$ is not used in estimating CM (except for those cases where there is no slam-induced whipping, e.g., barges, SWATHs, offshore platforms).

SLAM-INDUCED LOADS

Slam-induced whipping is an inherently nonlinear phenomenon for vertical and lateral bending due to the following factors:

1. The bow can emerge completely from the water and then the bottom impacts the sea (bottom slamming).
2. Alternatively, the water can rush up the bow and impact a flared portion of the bow (flare slamming). In either case, the slam-induced impact causes the hull girder to vibrate in its fundamental bending mode (vertical and/or lateral bending). This vibratory bending moment ($W_{P\text{-}P}$) is superimposed on the low-frequency seaway bending (M) and dampens out after a few wave cycles.

For each of those cells in which slamming may occur, a high frequency (quasi-dynamic) whipping load ($W_{P\text{-}P}$) is developed. Whipping peaks follow an exponential distribution whose characteristic values and truncation values are functions of ship geometry, operations, and sea states. In this method, the maximum whipping moment in a cell is truncated at the 1% probability of exceedance (if there are more than 100 slam events). In computing the maxi-

mum combined moment, the maximum whipping ($W_{P\text{-}P}$) moment is reduced by a small amount to account for the damping that occurs between slam initiation and the peak in the hog and sag moments and a nonlinear sag offset. The net whipping moment (W_i) for each cell is then combined with the maximum wave-induced moment of that cell to form a maximum combined moment (CM_i). The combined moment distribution for each cell is linear in semi-log space (on cycles) between CM_i (at one cycle) and the Rayleigh distribution of wave-induced loads at Nw_i, the number of whipping cycles. This is only an interim approach until that time when the form of the probability distribution of the combined wave plus whipping moment becomes known.

In this method, the maximum lifetime combined moment (CML) from all of the cells follows a numerical iteration procedure such that CML is exceeded by one cycle cumulatively. Specifically, each of the linear distributions for the whipping cells is extrapolated to various values of CML until a total of one cycle is exceeded. This is analogous to that previously described for M_{max_L}.

Since the cumulative probability of CML is not readily calculable, the following method is used for extreme loads. When the desired load M is greater than CM_i in the worst cell, the cumulative probability for each cell results from:

$$Y = (M - W_i)^2 / E_i \tag{8}$$

and

$$CP_i = [1 - \exp(-Y)]^{Ni} \tag{9}$$

Thus, the wave-induced portion of the combined load increases as a Rayleigh distribution, while the maximum whipping contribution is limited by the truncated exponential distribution. This procedure includes the effects from both speed and motion induced slamming. However, as more accurate probability distributions for the combined loads become available, the above equations can be readily modified.

As with the wave-induced loads, the net cumulative probability for load M for all of the cells is the product of the individual cumulative probabilities for each cell.

$$CP_{net} = \Pi_{i=1,n}\, CP_i \tag{10}$$

Hence, one can iterate among values of M to arrive at any desired occurrence probability, cumulative probability, or reliability.

COMBINED BENDING MOMENT

Slam events occur randomly throughout certain cells and not at all in other cells. The total load is a combination of the low frequency, wave-induced load with a high frequency whipping component added on. Since there may only be several hundred cells with slamming, it is feasible to determine the probability density functions from tests or analytical computer programs. However, this is still a major undertaking at the current state-of-the-art. An alternative is to employ empirical based algorithms to determine the probability density functions for the high frequency whipping component of each cell. The whipping distribution may then be combined with the low frequency portion from the Rayleigh distribution for that cell. This results in a numerical probability density function for each cell that is readily useable for reliability calculations.

The total bending moment (either vertical or lateral) following a slam event consists of the slam-induced moment superimposed on the wave-induced moment. Normally, the slam event occurs between the peak in the wave hog cycle and the peak in the sag cycle and is referred to as the phase angle (Φ). Several whipping cycles may occur before the sag portion of the whipping moment coincides with the peak wave sag moment. By following the peak in the wave hog cycle, the whipping moment is significantly reduced from hydrodynamic damping. The peak-to-peak variation of the combined bending moment (CM) is decidedly larger in the sag direction. The following equations show how the wave-induced vertical moments (M_{Hog} and M_{Sag}) and the lateral bending moments (amplitude moment = CM_L) combine with the peak-to-peak slam moments ($W_{P\text{-}P}$):

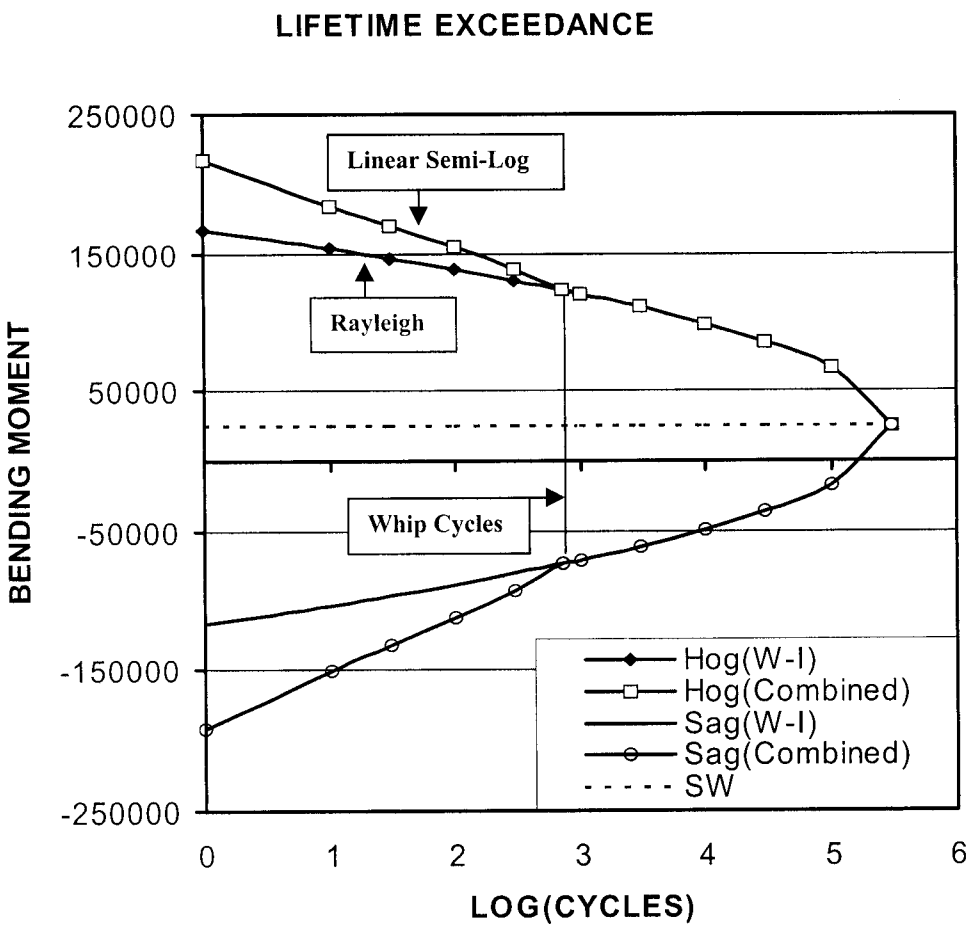


FIGURE 2: *Sample Load Exceedance Distribution for a Cell*

$$CM_{Hog} = M_{Hog} + SW + 0.5\ W_{P\text{-}P} \exp[\ -\delta R\ (\Phi + 180)/360] \quad (11)$$

$$CM_{Sag} = M_{Sag} + SW + 0.75\ W_{P\text{-}P} \exp[\ -\delta R\ (\Phi)/360] \quad (12)$$

$$CM_L = M_L + 0.5\ W_{P\text{-}P} \quad (13)$$

where SW is the still-water bending moment, R is the ratio of the natural frequency to the wave encounter frequency, and δ is the log decrement of the hydrodynamic damping of the hull. The initial whipping moment is biased in the sag direction.

The probability distribution for the total combined moment (vertical and lateral bending) is approximated in semi-log space (the X axis is the Log_{10} of the cumulative cycles in a cell and the Y axis is the combined load). The distribution is linear between the maximum combined lifetime load (CM) at one cycle and the number of slam events (NW_i) on the Rayleigh distribution from the wave-induced loads. The distribution follows the Rayleigh distribution for all cycles beyond (NW_i). **Figure 2** is an example of these distributions for a typical cell. The proposed linear distribution (in semi-log space) of the combined loads is an interim solution. As more data becomes available, the actual distribution or family of distributions may be used.

The lifetime exceedance load spectrum is obtained by selecting a load, checking each distribution to find the number of cycles that exceeds that load, and then summing the

Table 1

Operational Profiles for Combatants

		SIGNIFICANT WAVE HEIGHT (METERS)		
SPEED	*HEADING*	*0 - 5*	*6 - 10*	*> 10*
5	0	0.0410	0.0760	0.0600
5	45	0.0820	0.0840	0.7600
5	90	0.0820	0.0275	0.0000
5	135	0.0820	0.0540	0.0300
5	180	0.0410	0.0330	0.0000
15	0	0.0710	0.1160	0.0000
15	45	0.1430	0.2870	0.0920
15	90	0.1430	0.0455	0.0000
15	135	0.1430	0.1150	0.0080
15	180	0.0710	0.0600	0.0500
25	0	0.0130	0.0070	0.0000
25	45	0.0250	0.0190	0.0000
25	90	0.0250	0.0190	0.0000
25	135	0.0250	0.0190	0.0000
25	180	0.0130	0.0380	0.0000

Table 2

Operational Profiles for Amphibious or Fast Cargo Ships

		SIGNIFICANT WAVE HEIGHT (METERS)		
SPEED	*HEADING*	*0 - 5*	*6 - 10*	*> 10*
5	0	0.0380	0.0480	0.2290
5	45	0.0750	0.0820	0.2570
5	90	0.0750	0.0550	0.0845
5	135	0.0750	0.0670	0.1140
5	180	0.0380	0.0340	0.0570
15	0	0.0850	0.1520	0.0570
15	45	0.1700	0.2100	0.0860
15	90	0.1700	0.1240	0.0295
15	135	0.1700	0.1380	0.0570
15	180	0.0850	0.0700	0.0290
25	0	0.0020	0.0030	0.0000
25	45	0.0050	0.0050	0.0000
25	90	0.0050	0.0050	0.0000
25	135	0.0050	0.0050	0.0000
25	180	0.0020	0.0020	0.0000

cycles for that load. This procedure is repeated for an entire of set of loads and generates the lifetime load spectrum used for fatigue analyses. Through an iteration process, one can obtain the load that is exceeded by all of the cells exactly one time. This is the maximum lifetime load (that is, the most probable extreme value) and is exceeded 63.2% of the time. As an example, if 100 ships spent there entire lives in the same conditions, then 63 would each see one bending moment larger than the predicted lifetime maximum moment. However, since the distributions for each cell are defined, one can select any given load and calculate its reliability. Hence, through iterative means, one can calculate the load corresponding to any desired level of reliability.

Assessment of Cumulative Lifetime Loads

The methodology presented in this paper was embodied in a Windows-based computer program, SPECTRA. This section describes this methodology, input needed to use it, and its results. A user's manual for SPECTRA is available (Michaelson 2000). The methodology was constructed using a frequency domain approach to calculate the low frequency, wave-induced portion of the seaway loads. The high frequency, slam-induced whipping distribution (assumed to follow the exponential distribution) is developed from empirical algorithms for a variety of ship types and bow forms. This meets the goal of providing a consistent method for extreme loadings as well as a full exceedance loading distribution for fatigue. The data used to develop the empirical algorithms and the specific equations have been described by Sikora (1983, 1986, and 1998). The subsequent sections provide brief discussions of some of the important input parameters and random variables, and results from the methodology.

LIFETIME AT SEA

The lifetime at sea, in days, must be selected. Historically, Navy ships were only operated about one third of the time. However, with the reduced size of the active Fleet, ships may be operated two thirds of the time in the future. This includes down time for pierside re-supply, overhauls, ship alts, and repairs. Hence, for a 30-year life, a ship would be expected to spend 7,300 days at sea.

OPERATIONAL PROFILE

Ship Heading

The ship heading must be input for each RAO. SPECTRA uses units of degrees with head seas corresponding to zero degrees. Port/starboard symmetry can be assumed for the RAOs so one can input headings from 0° to 180° and double the percentage of time in which that heading occurs. For vertical bending, head seas, bow seas (45°), quarter following seas (135°), and following seas (180°) are sufficient (beam sea responses are virtually zero and can be ignored). However, for lateral bending and torsion a finer break-

down is needed; namely, head seas, bow seas (30° and 60° — the most critical heading), beam seas (90°), quarter following seas (120° and 150°), and following seas (180°) are all needed. Bow and quarter following seas can be frequently assumed to be equivalent, as can head and following seas.

Heading Probability

The probability (Ph_i) that the ship will operate at a given heading relative to the waves as functions of sea state must be input. Normally, one would expect the ship to be unaffected by low seas and hence to have random heading probabilities. As the seas become more severe, the ship likely will change heading to minimize certain motions and accelerations. The user will want to take this into account as well as any mission specific reasons for biasing heading probabilities. SPECTRA has a number of options for heading probabilities (for various types of ships) that the user can select from a list. **Tables 1** through **4** represent the results of a statistical study of actual Navy ship operations.

Ship Speed

The ship speed must be input for each RAO; SPECTRA requires units in knots. For commercial ships and naval auxiliaries, two speed ranges are usually sufficient: (1) Under 10 knots for rough weather, this can be idealized as 5 knots, and (2) the ship's design (cruise) speed. Combatants will likely need at least three speed ranges: (1) Under 10 knots for ASW missions and rough weather, this can be idealized as 5 knots, (2) a task force or convoy speed of 15 knots, and (3) the top operational speed. Special mission requirements may dictate a finer breakdown of speed for particular vessels. Aircraft carriers and high-speed commercial ships will require the same three speed ranges as combatants but with different probabilities of operations.

Speed Probability

The probability (Ps_i) that the ship will operate at a given speed relative to the waves as functions of sea state must be input. Normally, one would expect the ship speed to be unaffected by low seas and the design requirements to govern. As the seas become more severe the ship likely will slow down, and, in truly severe conditions it will just maintain enough speed for steerage. SPECTRA also has a list of options for speed probabilities for various types of ships that can be selected. Tables 1-4 present the joint heading and speed probabilities for several ship types as functions of sea state ranges.

Table 3

Operational Profiles for Aircraft Carriers

		SIGNIFICANT WAVE HEIGHT (METERS)		
SPEED	*HEADING*	*0 - 5*	*6 - 10*	*> 10*
5	0	0.0500	0.0480	0.1500
5	45	0.1030	0.1960	0.3680
5	90	0.1030	0.1600	0.0560
5	135	0.1030	0.0940	0.0800
5	180	0.0500	0.0300	0.0120
15	0	0.0560	0.0470	0.0750
15	45	0.1130	0.1180	0.1150
15	90	0.1130	0.1140	0.0280
15	135	0.1130	0.0720	0.0780
15	180	0.0560	0.0250	0.0380
25	0	0.0180	0.0080	0.0000
25	45	0.0350	0.0150	0.0000
25	90	0.0350	0.0290	0.0000
25	135	0.0350	0.0360	0.0000
25	180	0.0180	0.0080	0.0000

Table 4

Operational Profiles for Auxiliary or Slow Cargo Ships

		SIGNIFICANT WAVE HEIGHT (METERS)		
SPEED	*HEADING*	*0 - 5*	*6 - 10*	*> 10*
5	0	0.0250	0.0350	0.1750
5	45	0.0500	0.0750	0.1750
5	90	0.0500	0.0510	0.0870
5	135	0.0500	0.0420	0.1750
5	180	0.0250	0.0130	0.0880
15	0	0.1000	0.1490	0.0750
15	45	0.2000	0.2000	0.0750
15	90	0.2000	0.1850	0.0370
15	135	0.2000	0.1750	0.0750
15	180	0.1000	0.0750	0.0380

SEA STATE PROBABILITY

The probability (Pwh_i) that a given sea state (in one meter intervals) will occur must be selected. This is a function of the geographic area where the vessel is expected to operate. Lee (1995) compiled ocean and coastal statistics for the whole world. A study of actual Navy ship operations for the Atlantic Fleet showed that the Navy operates mostly in U.S. coastal waters, the Caribbean, Mediterranean,

Table 5

Wave Height Probability of Occurrence

WAVE HEIGHT (METERS)	STATION I (OCHI, 1978)	WALDEN DATA (ABS)	NATO NORTH ATLANTIC	GRID PT. 10 SOWM (LEE, 1995)	REALISTIC NAVY, NORTH ATLANTIC*
0 - 1	0.0503	0.04064	0.087	0.0953	0.36153
1 - 2	0.2665	0.28779	0.192	0.1889	0.32747
2 - 3	0.2603	0.30101	0.220	0.1906	0.16092
3 - 4	0.1757	0.18507	0.157	0.1600	0.07266
4 - 5	0.1014	0.08966	0.124	0.1262	0.03696
5 - 6	0.0589	0.03853	0.080	0.0902	0.01894
6 - 7	0.0346	0.02545	0.052	0.0532	0.00956
7 - 8	0.0209	0.01352	0.039	0.0380	0.00548
8 - 9	0.0120	0.01006	0.025	0.0253	0.00304
9 - 10	0.0079	0.00662	0.013	0.0164	0.00172
10 - 11	0.0054	0.00042	0.007	0.0088	0.00091
11- 12	0.0029	0.00035	0.004	0.0044	0.00052
12- 13	0.0016	0.00046	0.000	0.0015	0.00017
13 - 14	0.00074	0.00022	0.000	0.0008	0.00009
14 - 15	0.00045	0.00013	0.000	0.0005	0.00005
> 15	0.0004	0.00008	0.000	0.0000	0.00000

* *The realistic North Atlantic data were based on weighting the average time U.S. Navy ships spent in the North Atlantic, Caribbean, Mediterranean, and U.S. coastal waters. In actual practice, U.S. Navy ships avoided storms so the probabilities listed are conservative.*

and the Persian Gulf environs — all relatively benign areas. Time in the open ocean is mostly spent in transit to and from the Mediterranean and the Persian Gulf. However, for design purposes, a severe portion of the ocean is more appropriate (Station I in the North Atlantic (Ochi 1978) has been used for recent Navy designs). This is consistent with the American Bureau of Shipping's approach of requiring open ocean statistics (normally, the Walden North Atlantic data) for classing a ship for unlimited ocean operations. A comparison of wave height probabilities from several sources is presented in **Table 5**. (see caption correction) SPECTRA has a list of several wave height probabilities that can be selected as options if the user does not want to input his own data.

SEA SPECTRA

Sea spectra are described by functions of wave energy in the frequency domain. The simplest is the Pierson-Moskowitz (Pierson 1964), which, for a given wave height, gives energy as a function of wave frequency. This corresponds to fully developed seas and approximates the most probable modal periods that would be encountered in the open ocean (Pmp_i = 1.0). The Bretschneider spectrum (Bretschneider 1959) is a family of wave energy functions for a given wave height. Each function in the family is defined by the modal period. In SPECTRA, their are two versions of the Bretschneider spectra: one, where the user provides the modal period and their probabilities of occurrence (Pmp_i), and two, where the modal periods and their associated probabilities (Pmp_i) are determined from algorithms (Ochi 1978). A more sophisticated sea spectra was developed by Ochi (1978); it has a family of eleven functions for a given wave height. The modal periods and their associated probabilities (Pmp_i) are determined from algorithms embedded within SPECTRA. The Ochi spectra is recommended because it best represents the North Atlantic data.

MODAL PERIODS

If the user of SPECTRA chooses Pierson-Moskowitz, Ochi, or the second Bretschneider version, the wave periods and probabilities of occurrence or completely transparent to the user, i.e., they are internally computed from algorithms. However, the user may select his own family of wave periods and probabilities of occurrence. Lee (1995) includes this data for areas covering the whole world. ABS rec-

ommends the Walden data (Walden 1964) for the North Atlantic. In general, the longer wave periods are more prevalent from the Ochi algorithms and the shorter ones more likely from the Walden data. However, there is relatively little difference for the most critical waves for naval ships (i.e., wave heights between 11 and 15 meters).

RESPONSE AMPLITUDE OPERATORS

SPECTRA requires RAOs for each combination of heading and speed. These may either

Table 6

Speed Factor Coefficients

SHIP TYPE	SPEED COEFFICIENT (C_{SPEED})
Commercial Ships & Naval Auxiliaries	0.0075
Amphibious Assault	0.025
Aircraft Carriers	0.006
Frigates	0.011
DD 963 & CG 47	0.007
DDG 51	0.009
Wave-piercing Bow Variant of DD 21	-0.010
Other Destroyers & Cruisers	0.011

Table 7

Universal Square Root RAOs

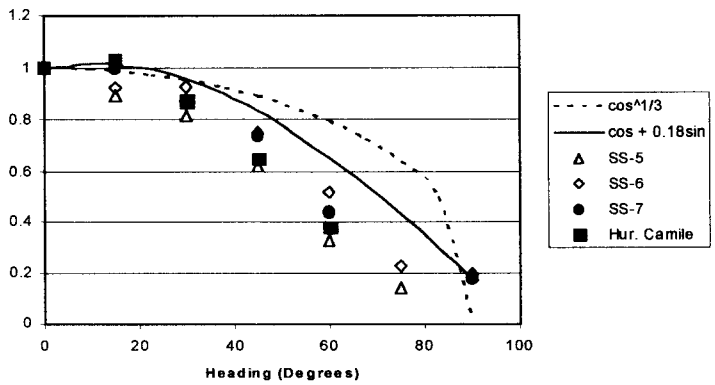
NORMALIZED WAVE FREQUENCY $F_3\,\Omega$	NORMALIZED ROOT RAO $\sqrt{\text{RAO}}\ D\ F_1 F_2$	NORMALIZED WAVE FREQUENCY $F_3\,\Omega$	NORMALIZED ROOT RAO $\sqrt{\text{RAO}}\ D\ F_1 F_2$
0.4	0.0	1.1	0.0167
0.5	0.0018	1.2	0.0143
0.6	0.0045	1.3	0.0122
0.7	0.0078	1.4	0.0100
0.8	0.0114	1.5	0.0080
0.9	0.0151	1.6	0.0064
0.95	0.0170	1.7	0.0051
1.0	0.0180	1.8	0.0042
1.05	0.0177	2.0	0.0036

Table 8

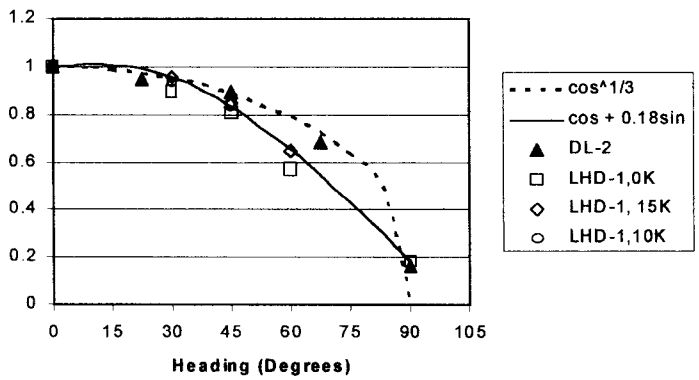
Summary of Square Root RAO Factors

FACTOR	VERTICAL BENDING MOMENT FACTORS	LATERAL BENDING MOMENT FACTORS	TORSIONAL MOMENT FACTORS
D Dimensional Normalization Factor	$\rho g L^2 B$	$\rho g L^2 \sqrt{BT} /(C_b+0.25)$	$C_T \rho g L^2 B^2 / T$
F1 Heading Factor for Magnitude	$\cos(\theta)+ 0.18 \sin(\theta)$	$0.28 + 0.73 \sin (1.64\ \theta)^{1.5}$	$[3-2\cos(X)+\theta(1+\cos(X))/360]/5$
F2 Speed Factor for Magnitude	$1.0 + C_{speed}\, v$	$1.0 + C_{speed}\, v$	$1.0 + C_{speed}\, v$
F3 Heading Factor for Wave Frequency	$(\cos(\theta))^{0.5}$	$\cos(\theta) + 0.275\ (\sin(\theta))^2$	$1.0 - 0.58 \tanh[1.0 - ((90 - \theta)/90)^2]$

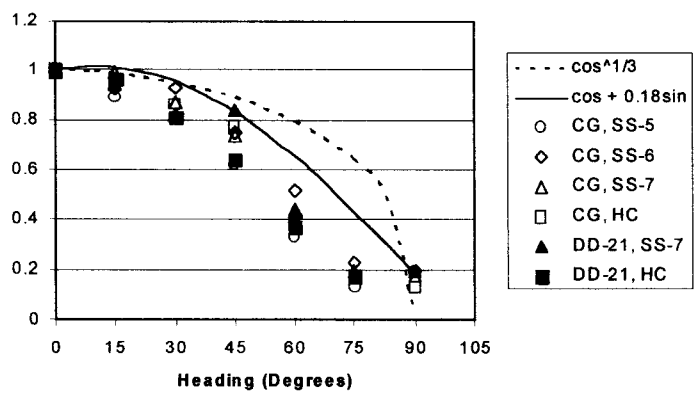
Where: *L = ship length between perpendiculars*
B = maximum beam at waterline
T = mean draft
C_b = block coefficient
C_{speed} = speed coefficient (Table 1)
C_T = torsional coefficient (0.0313 for containerships and 0.043 for closed ships)
ρg = density of seawater (e.g., 0.0286 ltons/ft3)
$X = \theta^2 / 5 \ln (\theta + 1.001)$
θ = ship heading (degrees), 0° = head seas
v = ship speed (knots)
g = gravity acceleration (e.g., 32.2 feet/sec², 19.2 knots/sec)



a. Land Attack Destroyer



b. Amphibious Support Ships



c. Cruisers and Wave-piercing Bow Combatants

FIGURE 3:
Relative Magnitude of Vertical Bending Moments as Functions of Heading

be user input or from internal algorithms. Sea trials and model tests have shown the load RAOs to be independent of sea state. However, if this is found not to be the case for a particular vessel, one can input two RAOs for the same speed and heading and adjust the probability of occurrence of those conditions for different sea states.

The user input RAOs must be in the wave frequency domain (not the wave encounter domain). The corresponding heading and speed must also be indicated for each RAO. Simple strip theory codes usually provide reasonable vertical bending moment RAOs for head seas at zero speed but are inconsistent at forward speed. One can use the heading and speed algorithms (Sikora, 1998) to manually generate the rest of the RAOs from the head sea, zero speed case. Some of the three-dimensional codes can generate vertical and lateral bending moment RAOs in oblique seas and at speeds. However, a full validation of these codes remains to be done.

SPECTRA has internal algorithms for generating RAOs (for conventional monohulls only) for vertical bending, lateral bending, and torsion. These algorithms are empirically based and are functions of ship length, beam, and draft, as well as speed and heading. **Tables 6** through **8** summarize these algorithms, which were originally presented by Sikora (1983, 1998). However, some recent analysis of model test data has led to a proposed revision of the heading factor for vertical bending. **Figure 3** shows example comparisons of the ratio of wave-induced vertical bending moments at bow through beam sea headings relative to head seas. The original algorithm, $F_l = (cos(\theta))^{1/3}$, was based on model test data of the DL 2 destroyer. Model test data from the LHD 1, the DDG 51, the CG 61, and a wave-piercing bow variant of the DD 21 showed it to be overly conservative in the 30° to 80° range of headings. Therefore, a revised algorithm is proposed:

$$F_l = \cos(\theta) + 0.18 \sin(\theta) \qquad (14)$$

The algorithms for vertical bending of destroyers, frigates, and cruisers have a net uncertainty of 0.969 for the mean (i.e., the

algorithm is slightly conservative) and a standard deviation of 0.168. Aircraft carriers, amphibious support, and high-speed cargo ships have net uncertainties of 0.962 for the mean and 0.146 for the standard deviation. There is insufficient data for slow speed commercial ships or for lateral bending to calculate uncertainties.

The algorithms assume a one minus cosine distribution along the length of the ship with the maximum value at midships. However, for design purposes, one may choose to use the midships value for stations 8 through 11 to obtain a more robust design.

WHIPPING

Algorithms for generating the Weibull parameters (characteristic values, truncation values, and slope equal to 1.0 for the exponential distribution) for slam-induced whipping for each cell are contained within SPECTRA. The ship length, beam, and displacement are required inputs while the ship's speed, heading, and wave heights are defined by the particulars of the cells. The user also selects ship type and bow form from drag down lists; this allows SPECTRA to decide which cells would have whipping.

The whipping moments for each cell are determined from an exponential distribution, which requires the characteristic value (Chv) and truncation value (Xo). The following algorithms were developed from extensive model test and sea trial measurements on the CG 61 and LHD 1 as well as smaller data samples of several other ships. The algorithm for the peak-to-peak vertical bending moments from slam-induced whipping is:

$$Chv_{vwhip} = C_{bow}\, F_2\, F_H\, L^2\, B \cos(\theta) \qquad (15)$$

where C_{bow} is a bow form coefficient, F_H is a wave height factor, and F_2 is the same speed factor used for the RAOs. **Table 9** presents some of the best-fit values of C_{bow} for several ship types.

A wave height factor (F_H) for the largest waves was developed:

$$F_H = 1.0 \quad \text{for } H < 8\text{m} \qquad (16a)$$

Table 9

Bow Form Coefficients

BOW FORM	C_{bow}	CL_{bow}
Fine Bow Frigates - V Bow	0.00092	0.00133
Fine Bow Frigates/Destroyer - Bow Dome	0.00120	0.00133
Flared Bow Cruiser - Bow Dome	0.00193	0.00180
Flat Bottom Cargo	0.00200	0.00200
Large Bow Flare Amphibious	0.00270	0.00240
Large Bow Flare Aircraft Carrier	0.00345	0.00200

$$F_H = 1.0 + 0.0000846\, L\, (H - 8) \quad \text{for } H > 8\text{m} \qquad (16b)$$

where H is the significant wave height (meters) and L is the ship length between perpendiculars (feet).

The algorithms for lateral bending moments from slam-induced whipping are given by:

$$Chv_{Lwhip} = CL_{bow}\, F_2\, L^2\, B\, F_\theta \qquad (17)$$

where CL_{bow} is a bow form coefficient, F_2 is the same speed and wave height factor used for vertical bending, and F_θ is a heading factor as given by:

$$F_\theta = 0.5\,(\sin\theta)^3 + 0.87\,(\cos\theta)^{1/3} \qquad (18)$$

The truncation value for vertical and lateral bending for slam-induced whipping is given by:

$$Xo = 0.001\, C_{bow}\, L^3\, B \qquad (19)$$

Sikora (1998) shows the agreement of these algorithms with at sea measurements.

For the exponential distribution, the maximum peak-to-peak whipping moment ($W_{P\text{-}P}$) in a cell i is:

$$W_{P\text{-}P} = Chv_{whip} \ln(nw_i) + Xo \qquad (20)$$

where Nw_i is the number of slam events in the cell. Even though the model data show the exponential distribution, extreme whipping responses are unlikely to be seen by ships at sea. Ship operators reduce speed and change heading to avoid the accelerations from severe

whipping. Therefore, an upper limit can be imposed on the exponential distribution at a one percent probability of exceedance. This corresponds to a value 4.6 times the characteristic value plus the truncation value.

The uncertainty associated with the characteristic values of the whipping moments was determined from the empirical database used to generate the algorithms. The uncertainty distribution of the data is normal with a mean value of 97.05% of the algorithm (i.e., the algorithm is slightly conservative) and a standard deviation of 24.65%.

A trapezoidal distribution, along the length of the ship, for the whipping moments is assumed. It uses the midships values from Station 8 to Station 10 for vertical bending (and Station 8 to Station 12 for lateral bending) and drops to zero at the bow and stern.

SLAM EVENTS

The number of slams in a particular cell can be calculated entirely within SPECTRA, or the user can enter a value for the average time between slams. In order to determine the number of slam events in a cell, the slamming rate must be known. For headings, speeds, and sea conditions in which significant slamming is expected to occur, the slam rate is assumed to be a function of the encounter frequency. The encounter frequency (f_e) in wave encounters per hour can be expressed as:

$$f_e = 720/\sqrt{H} + 46.8\ (v/H)\cos(\theta) \qquad (21)$$

where H is in meters and v is in knots. This implies that head seas are the most severe heading; the slamming rate increases with speed; and that as the wave heights increase, the wave periods increase, and the slamming rate decreases. The following algorithms for estimating slam rates were developed as functions of the encounter frequency and ship length:

$$\text{Slam Rate (Bow flare)} = 72 f_e / L \qquad (22)$$

$$\text{Slam Rate (Fine bow)} = 30 f_e / L \qquad (23)$$

These slam rate algorithms are reasonable for both vertical and lateral slamming. When a ship experiences an impulsive event, both vertical bending and lateral bending are usually excited. Sikora (1998) shows the agreement of the algorithms with measured rates at sea.

COMBINED MOMENTS

The combination of whipping and wave-induced bending moment components is a function of the natural frequency (lowest bending mode), phase angle Φ (the average time in the wave cycle when slamming occurs), and log decrement of damping δ. The user can override the default algorithms by entering any or all of these parameters. The longitudinal distribution of the combined moments is based upon summing the one minus cosine distribution for the wave-induced portion and the trapezoidal distribution for the slam-induced portion.

RINGING

When a ship is struck by an impulsive slamming force, it vibrates as a free-free beam at its natural frequency until it is damped out; this is called ringing. The initial whipping moment was combined with the peak wave moment but the subsequent ringing cycles were not included. These ringing cycles contribute to the overall exceedance loads for fatigue but not to the extreme load calculations for reliability. All of the subsequent ringing cycles for each slam event are assumed to vary about a zero mean and are added into the load exceedance spectra. The net effect is a very slight increase in the fatigue damage from ringing. The natural frequency of the ship can be input by the user or it can be calculated within SPECTRA. Sikora (1998) presents algorithms for predicting these frequencies.

STILLWATER BENDING MOMENT

The still-water bending moment is a significant portion of the total vertical bending moment. The still-water moment can be generated analytically; however, the hull form and weight distribution must be known. The Ship Hull Characteristic Program (SHCP) is a static balance approach that is capable of predicting stillwater bending moments and

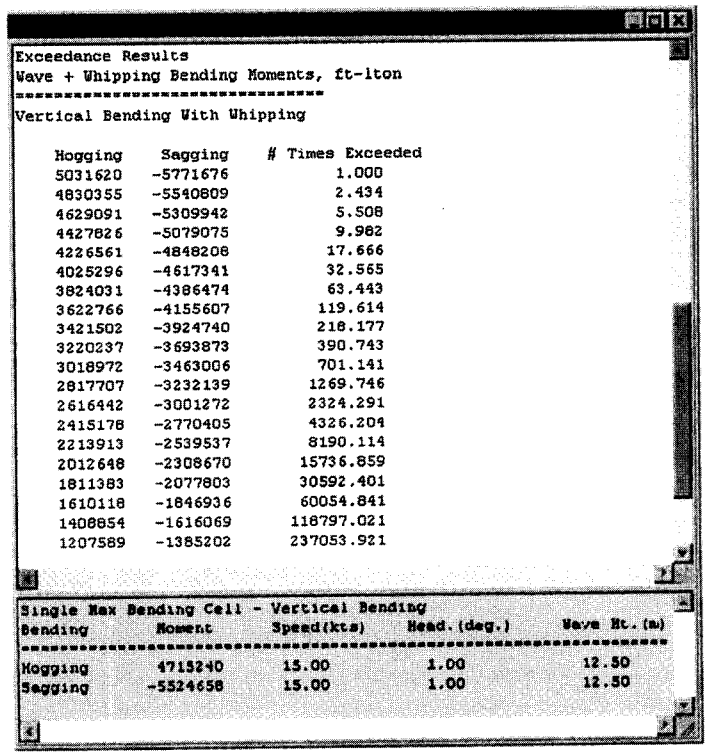
vertical bending moments from a static balance on a wave profile. It is easy to use and is the industry standard tool for hydrostatics. In this methodology, SHCP is recommended for calculating stillwater bending moments.

If one enters a zero stillwater moment in the SPECTRA program, then all of the bending moments will be relative to the stillwater moment. On the other hand, if the actual stillwater moment is used, then the loads will be absolute.

Summary

The methodology is described in detail by Sikora (1998), as it is embodied in the Windows-based computer program, SPECTRA (Michaelson, 2000). The following variables comprise the input required to use the methodology:

1. User supplied input
   - The name of the ship or test case The length between perpendiculars (English or metric units)
   - The beam at midships (English or metric units)
   - The nominal draft (English or metric units)
   - The displacement (English or metric units)
   - The location along the length of the ship where the load is to be calculated
   - The stillwater bending moment at the location of interest
2. Selections from pull down menus:
   - Type of load (vertical bending, lateral bending, or torsion)
   - Ship type (destroyer, auxiliary, aircraft carrier, etc.)
   - Bow form (fine bow, destroyer with bow flare, carrier bow, etc.)
   - Sea spectra (Ochi six-parameter, Pierson-Moskowitz, etc.)
3. Choice from pull down menu or user supplied from external data file:
   - Probability of wave heights (North Atlantic Station I, NATO Atlantic)
   - Operational profile (destroyer, carrier, etc.)
   - RAOs (one for each heading and speed combination)



Exceedance Results
Wave + Whipping Bending Moments, ft-lton
Vertical Bending With Whipping

Hogging	Sagging	# Times Exceeded
5031620	-5771676	1.000
4830355	-5540809	2.434
4629091	-5309942	5.508
4427826	-5079075	9.982
4226561	-4848208	17.666
4025296	-4617341	32.565
3824031	-4386474	63.443
3622766	-4155607	119.614
3421502	-3924740	218.177
3220237	-3693873	390.743
3018972	-3463006	701.141
2817707	-3232139	1269.746
2616442	-3001272	2324.291
2415178	-2770405	4326.204
2213913	-2539537	8190.114
2012648	-2308670	15736.859
1811383	-2077803	30592.401
1610118	-1846936	60054.841
1408854	-1616069	118797.021
1207589	-1385202	237053.921

Single Max Bending Cell - Vertical Bending

Bending	Moment	Speed(kts)	Head.(deg.)	Wave Ht.(m)
Hogging	4715240	15.00	1.00	12.50
Sagging	-5524658	15.00	1.00	12.50

FIGURE 4: *SPECTRA Output for Graphical Exceedance Plots*

4. Choice of default algorithm within SPECTRA or user entered on input screen:
   - Whipping frequency (units of Hertz)
   - Log decrement of hydrodynamic damping
   - The average time between slam events (units of seconds)
   - The phase angle between wave cycle and slam initiation (units of degrees)

The output from SPECTRA consists of graphs (**Figure 4**) and tables (**Figure 5**) of load (25 hogging and sagging values for vertical bending, or 25 one-sided amplitudes for lateral bending and torsion) with the number of times the load is exceeded during the lifetime of the ship. This is given for both the wave-induced loads alone and for the combination of the wave-induced and whipping. The maximum lifetime load is that value exceeding one cycle.

Fatigue histograms (and graphs) are also generated for 24 mean loads and the number of lifetime cycles occurring at each load. These data are also provided for both wave-induced only and for the combined wave-induced with whipping.

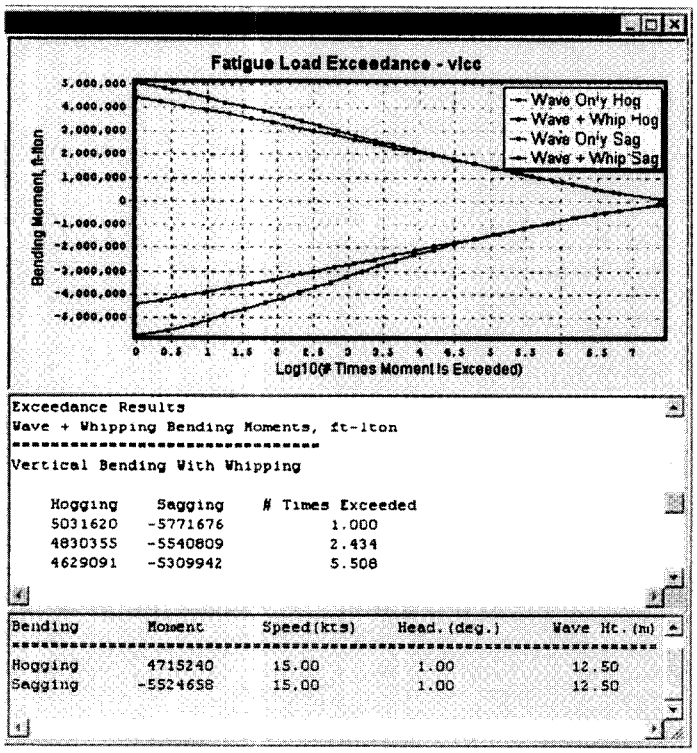


FIGURE 5:
SPECTRA Output in Text Format

The reliability corresponding to the maximum lifetime load is presented along with reliabilities at 1.1, 1.2, 1.3, and 1.4 times the maximum lifetime load. The user can interpolate the reliabilities of other loads by assuming a linear variation of load with the Log_{10} of the reliabilities.

Acknowledgements

This report represents the current stage of an evolving process initiated at Carderock during the 1960's. The pioneering works of Mr. Jack Birmingham, Mr. John Andrews, and most of all, Mr. Alfred Dinsenbacher are gratefully acknowledged. During the intervening decades, the following colleagues made substantial contributions to this process: Dr. Jeffrey Beach, Mr. Thomas Brady, Mr. John Dalzell, Mr. William Hay, Dr. David Kihl, and Mr. Richard Lewis. Finally, the encouragement and financial support of Mr. Allen Engle, Mr. Jeffrey Hough, and Mr. Michael Sieve of NAVSEA 05 is greatly appreciated. ■

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PROFESSOR SHAHRAM SARKANI is a member of the engineering management and systems engineering faculty at The George Washington University (GW). He earned the B.S. (1980) and the M.S. (1981) degrees from Louisiana State University and his Ph.D. (1986) from Rice University, all in civil engineering. Professor Sarkani joined GW's faculty in 1986. He served as chair of the Civil, Mechanical, and Environmental Engineering Department from 1994 to 1997 and joined the School of Engineering and Applied Sciences (SEAS) Department of Systems Engineering and Engineering Management in 1998. Since 1997 he has been the SEAS Interim associate dean for Research and Development. He was awarded the Walter L. Huber Civil Engineering Research Prize by the American Society of Civil Engineers in 1999. The broad areas of stochastic methods of structural dynamics and fatigue, fatigue and fracture reliability, structural safety and reliability, and smart infrastructure systems for natural hazard mitigation constitute his research specialties. In 1993 he founded, and still directs, GW's Center for Infrastructure Safety and Reliability, whose main objective is to formulate and solve problems that require application of the most up-to-date and powerful theories of reliability and structural analysis. His current research investigations include modeling damage accumulation in materials made from advanced composites, modeling fatigue and fracture in welded joints, understanding extreme responses of structures to nonstationary vibratory loadings, using smart materials and dampers to control and reduce structural vibration, and testing large-scale structural models under stochastic loadings. Professor Sarkani's research activities are funded by the NSF, U.S. Navy, the National Aeronautics and Space Administration, and the U.S. Agency for International Development. His research has also been supported by the Department of the Interior, Department of Transportation, Office of Naval Research, and the National Institute of Standards and Technology. Professor Sarkani is the author of more than 100 technical publications and presentations.

MR. JEROME P. SIKORA has been with NSWCCD since he received his B.S. degree in physics from the University of Detroit in 1969. He is the head of the eighteen member Design Applications Branch (Code 6510), which is responsible for developing structural design criteria and design tools for Navy surface ships and submarines. In addition, it performs large-scale numerical analyses and experimental studies of whole ship and structural component behavior. Mr. Sikora began his career specializing in the field of static and dynamic holographic and other optical measurements of such structural parameters as stress, deformations, and vibration modes. Over the past two decades, he has developed probability-based load criteria for conventional and SWATH ships. He is the author of the lifetime, seaways load program, SPECTRA. Recently, he has been involved in developing structural design criteria and design methods for the cellular structure of no-frame, double hull ships; large floating structures, such as the Mobile Offshore Base; and high-speed sealift ships. He has written 100 government reports and published 30 refereed papers. ■